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REGULAR ARTICLE

The effect of materialism on unethical behaviour: The mediating role of self-control

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Abstract

Self-enhancement values (power, achievement, and hedonism) have been found to be important predictors of ethical behaviour; however, the link between materialism (also a kind of self-enhancement value) and unethical behaviour and its underlying mechanisms have received limited attention from researchers. The present research examines the effect of materialism on unethical behaviour and the mediating role of self-control across five multi-method studies with Chinese samples. Using Chinese data from the Sixth World Values Survey, we confirmed that materialism was positively related to a variety of unethical behaviours such as fare evasion, theft, tax evasion, bribery, domestic violence, and violence against others (Study 1). In follow-up experiments, priming materialism increased unethical behaviour (Studies 2 and 3). Moreover, through a questionnaire survey, we found that self-control played a mediating role between materialism and unethical behaviour in specific social contexts (Study 4). Finally, the manipulation of high materialists' self-control reduced their unethical behaviour, further verifying the mediating role of self-control (Study 5). Our findings indicate that materialists tend to have lower levels of self-control, thus they are more likely to engage in unethical behaviours. This research supports the hot/cool system model of self-control and deepens our understanding of how materialism might lead to unethical behaviour.

KEYWORDS
materialism, mediation, self-control, unethical behaviour

1 | INTRODUCTION

Unethical behaviour refers to behaviour that is illegal or morally unacceptable to the larger community, including violent and non-violent crime, delinquency, cheating, and other behaviours that violate ethical principles (Lu et al., 2020). It has brought about great negative impacts on our society's development and citizen's well-being. For instance, according to a recent survey report on global economic crime and fraud, nearly half of all people have encountered fraud in the past year, with a total loss of USD 42 billion (PwC, 2020). Therefore, it is important to find out the causes of unethical behaviour and then to fight against it based on scientific findings.

In recent years, there has been growing attention paid to the antecedents of unethical behaviour from researchers, and values have been suggested important psychological predictors of ethical behaviour. For instance, Feldman et al. (2015) demonstrated through a meta-analysis, a laboratory experiment, and a real-life event study that unethicality was positively associated with self-enhancement values (power, achievement, and hedonism) and negatively associated with self-transcendence (benevolence and universalism) and conservation (tradition, security,and conformity) values. Ring et al. (2020) examined the relationship between basic values and unethical behaviour in the sports field, and the results showed that self-enhancement values were positively

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related to doping likelihood, while self-transcendence values were negatively related to it. Arciniega et al. (2019) found that individuals who place relative importance on power values were more likely to exhibit unethical behaviour in the workplace.

Materialism, as a personal value which regards material wealth as the criterion for success, the center of life, and the source of happiness (Richins & Dawson, 1992), has also been proven to be a kind of self-enhancement value (Burroughs & Rindfleisch, 2002). Does it have a positive link with unethical behaviours? Some studies (e.g., Chowdhury & Fernando, 2013; Gentina et al., 2018; Koul, 2012; Liang et al., 2016; Lu & Lu, 2010) have provided supporting evidence for this question. However, the vast majority are correlational and the psychological mechanisms underlying the relationship between materialism and unethical behaviour need to be further explored.

According to a global survey performed in 2016 by Ipsos, a French research organization, 70% of Chinese, topping the list, believe that success can be measured through the things one owns (Ipsos, 2017). Considering that materialism is prevailing in China in recent years (Xu & Hamamura, 2014), the present research aims to explore the effect of materialism on unethical behaviour and the mediating role of self-control between them across five multi-method studies in Chinese samples. This research will help us understand how social values might negatively influence the societal well-being.

1.1 | Materialism and unethical behaviour

Materialists have been thought to be greedy and self-centered (Belk, 1985; Shrum et al., 2014). They regard material acquisition as the center of their lives and as their source of happiness and tend to judge one's success based on the number and quality of possessions accumulated (Richins & Dawson, 1992). Thus, materialists are more likely to engage in unethical behaviours for material gain, self-benefit, and recognition from others. A small number of studies have confirmed this. For example, Koul (2012) surveyed more than 2000 high school students in Thailand and found that materialism was positively associated with students' willingness to cheat and the frequency of self-reported cheating behaviour. Similarly, Gentina et al. (2018) found that when individuals use material acquisition rather than material sharing strategies to cope with loneliness, they exhibit more unethical behaviours. Chowdhury and Fernando (2013) surveyed nearly 1000 Australian consumers online via email and found a positive correlation between materialism and consumers' unethical beliefs, which is consistent with the findings of Lu and Lu's (2010) survey of Indonesian consumers. Overall, the vast majority of the existing research has been correlational. Only a very few studies have revealed the causal relationship between materialism and unethical behaviour. A study conducted by Liang et al. (2016) showed that trait materialism was positively related to corrupt intention, and state materialism induced experimentally could also increase corrupt intention. Therefore, the first aim of this research is to test whether the positive relationship between materialism and unethical behaviour reported in previous studies can be replicated in our study, and we propose Hypothesis 1: Materialism has a positive effect on unethical behaviour.

1.2 | Self-control as a possible mediator

Only very few studies have explored the mechanisms underlying the relationship between materialism and unethical behaviour. For instance, Du (2018) found that machiavellianism played a mediating role between the two. Another study revealed that the impact of materialism on corruption worked through citizens' life satisfaction, negative evaluations of corruption, and participation in public affairs (Peng, 2017). The present study suggests that self-control, an ability that can change an individual's internal reactions and inhibit undesirable behaviours (Tangney et al., 2004), may also be a potential mediating mechanism between materialism and unethical behaviour.

Metcalfe and Mischel (1999) proposed a hot/cool system for understanding the processes that enable and undermine self-control or willpower. In their framework, the cool system is cognitive, emotionally neutral, and contemplative, and the hot system is impulsive and passionate. The cool system is the seat of self-control and the hot system undermines efforts at self-control. de Ridder et al. (2012) believed that when the hot system is dominant, individuals exercise less conscious self-control and are therefore relatively prone to impulsive behaviour. Previous studies have demonstrated the positive effect of materialism on impulsiveness (Li et al., 2016; Roberts & Pirog, 2013; Rose, 2007). This suggests that to some extent, materialists' behaviour is guided by the hot system, which means they have lower levels of self-control. In addition, some studies also verified the negative relationship between materialism and self-control (Kim, 2013; Li et al., 2016).

Human beings may have a moral conscience, but they also have the more basic impulses to do what brings immediate benefit and to a certain extent, exhibit unethical behaviour (Baumeister & Alghamdi, 2015). Fortunately, self-control enables people to resist impulses and overcome selfishness so as to act in morally desirable ways (Gottfredson & Hirschi, 1990). Previous studies have demonstrated that self-control has a negative influence on unethical behaviour. For example, Through 32 years of longitudinal study, Moffitt et al. (2011) found that

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compared with children with high self-control, children with low self-control have a higher likelihood of being arrested for crimes in later life. Fan et al. (2016) revealed that participants with lower self-control had more deception behaviour and inclination. Other research also showed self-control decreased unethical behaviour (Fan et al., 2019; Muraven et al., 2006; Ren et al., 2018; Wang et al., 2017).

Based on the discussion, we propose Hypothesis 2: Self-control plays a mediating role in the relationship between materialism and unethical behaviour; specifically, materialists tend to have lower levels of self-control, thus they are more likely to engage in unethical behaviour.

1.3 | The current study

Although there has been supporting evidence for the positive link between materialism and unethical behaviour, the number of studies are still limited and the vast majority are correlational. Also, the findings of previous studies need to be replicated with multiple indicators of unethical behaviour in broader samples. In addition, the psychological mechanisms underlying the relationship between materialism and unethical behaviour need to be further explored.

Considering these limitations, the current research aims to examine the relationship between materialism and unethical behaviour and the mediating role of self-control in this relationship across 5 multi-method studies in Chinese samples. In Study 1, we examine the relationship between materialism and several unethical behaviours using the Chinese data from the Sixth World Values Survey (https://www.worldvaluessurvey.org/wvs.jsp). In Studies 2 and 3, we clarify the causality between materialism and unethical behaviour by manipulating materialism experimentally. In Study 4, we investigate the mediating role of self-control in the relationship between materialism and unethical behaviour through a questionnaire survey. Finally, in Study 5, we manipulate high materialists' self-control to examine the causal relationship between self-control and unethical behaviour to further verify the mediating role of self-control.

2 | STUDY 1

2.1 | Method

2.1.1 | Participants

The data were drawn from the WVS (Wave 6),[1] which included a total of 4538 participants recruited from China, aged between 18 and 85 years (M_{age}=44.51, SD_{age}=15.95), with 23 participants not reporting their age. Approximately 52.1% of the sample were female. A total of 2300 participants were from mainland China, 1238 from Taiwan, and 1000 from Hong Kong.

2.1.2 | Measures

Materialism

Following Gu et al. (2018), we used item V71 (It is important for this person to be rich and to have a lot of money and expensive things) to measure an individual's level of materialism, which has a 6-point response format ranging from 1 (very much like me) to 6 (not at all like me). After reverse-scoring the item, higher scores reflect higher levels of the individual's materialism.

Unethical behaviour

Following Peng (2017) and Xin and Li (2020) based on Chinese cultural background, we selected items V198, V199, V200, V201, V202, V208, and V210 to measure unethical behaviour. These items represent unjustified enrichment, fare evasion, theft, tax evasion, bribery, domestic violence, and violence against others, respectively. Participants rated these actions on a 10-point scale (1=never justified, 10=always justified), with higher scores representing higher levels of unethical behaviour.

2.2 | Results and discussion

As shown in Table 1, materialism was positively related to all variables (r=0.06–0.13, ps<0.001) except unjustified enrichment (r=−0.01, p=0.44). Regression analysis found that even after controlling for gender, age, and subjective social class, materialism could still significantly predict unethical behaviours such as fare evasion (B=0.08, p=0.001, 95% CI=[0.03, 0.13]), theft (B=0.09, p<0.001, 95% CI=[0.05, 0.12]), tax evasion (B=0.14, p<0.001, 95% CI=[0.09, 0.18]), bribery (B=0.15, p<0.001, 95% CI=[0.11, 0.18]), domestic violence (B=0.17, p<0.001, 95% CI=[0.12, 0.21]), and violence against others (B=0.17, p<0.001, 95% CI=[0.12, 0.22]).

Based on the database of WVS, Study 1 found that materialism positively predicted a variety of unethical behaviours, which was consistent with the findings of previous studies. However, Study 1 had several limitations. First, materialism was measured with a single item, which might not be able to fully reflect the characteristics of materialism. Second, the items tapping unethical behaviour did not directly measure unethical behaviour or intentions to engage in unethical behaviour. Third, Study 1 was still correlational research and could not reveal the causal relationship between materialism and unethical behaviour. Therefore, in Study 2, we used experimental methods to test this further. In addition,

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TABLE 1 Descriptive statistics and correlation analysis in Study 1.

	1	2	3	4	5	6	7	8	9	10	11
1. Gender[a]	1										
2. Age	0.00	1									
3. Subjective social class	−0.02	0.12***	1								
4. Materialism	−0.10***	−0.15***	0.00	1							
5. Unjustified enrichment	0.01	−0.05**	0.06***	−0.01	1						
6. Fare evasion	−0.02	−0.07***	0.00	0.06***	0.28***	1					
7. Theft	−0.04*	−0.09***	0.00	0.09***	0.19***	0.52***	1				
8. Tax evasion	−0.05**	−0.08***	−0.01	0.11***	0.13***	0.50***	0.64***	1			
9. Bribery	−0.07***	−0.07***	−0.01	0.13***	0.15***	0.43***	0.65***	0.69***	1		
10. Domestic violence	−0.14***	−0.03*	0.07***	0.13***	0.08***	0.28***	0.38***	0.38***	0.40***	1	
11. Violence against others	−0.07***	−0.10***	0.02	0.13***	0.08***	0.31***	0.37***	0.41***	0.39***	0.54***	1
M	1.52	44.51	3.44	3.87	4.18	2.60	1.69	2.09	1.86	2.05	2.23
SD	0.50	15.95	0.92	1.33	2.77	2.05	1.43	1.80	1.61	1.84	1.96

[a]Male = 1, female = 2.
*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

we used a more ecologically valid indicator of authentic behaviour as a measure of unethical behaviour, specifically examining individuals' unethical behaviour under monetary rewards.

3 | STUDY 2

3.1 | Method

3.1.1 | Participants

After obtaining the institutional ethical approval, a total of 64 undergraduates at a university in Hubei Province in China were recruited to participate in the experiment. After excluding five participants who did not follow the instructions of the experiment, the valid sample size was 59 (12 males; $M_{age} = 20.16$, $SD_{age} = 1.37$). All participants received a basic payment of 5 yuan and some additional rewards. The additional rewards depended on the participants' choice of the logical reasoning question.

3.1.2 | Experimental design and procedures

The experiment was a single-factor between-subjects design. After arriving at the laboratory, participants were randomly assigned to the materialism priming group and control groups. We then used an imagination task to manipulate participants' level of materialism. Subsequently, all participants completed manipulation checks using the state materialism scale. Finally, participants answered logical reasoning questions to measure unethical behaviour.

3.1.3 | Materials and measures

Materialism manipulation

Materialism was manipulated with the imagination-based writing task paradigm (Ku et al., 2014; Li et al., 2018). Participants in the materialism priming group wrote down the material goods they would like to own and the benefits of possessing those goods. Then they were asked to imagine receiving a considerable sum of money and to list what they would buy with that money. Participants in the control group were asked to list several feasible routes from the dormitory to the gate of the school, and then to choose one and explain why they chose that route.

State materialism

We used an 18-item material value scale (MVS) developed by Richins and Dawson (1992) and prefixed "at this moment" to the beginning of each item (e.g., At this moment, I admire people who own expensive homes, cars, and clothes.). All items were scored on a 5-point scale

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(1=strongly disagree, 5=strongly agree), and after reverse scoring, the total score was used to represent the level of individuals' state materialism. In this study, the Cronbach's α was 0.79.

Unethical behaviour

Unethical behaviour was measured with logical reasoning questions (Zhang, 2016). Participants were asked to answer four logical reasoning questions that were practically unsolvable (e.g., If 1=6, 2=16, 3=216, 4=2146, then 5=?). Then, participants were told that they had 4 min to complete the task and were allowed to choose an option between A (knowing the answer) and B (not knowing the answer). For each question they chose to "A. Knowing the answer", participants were rewarded with an additional 1 yuan in addition to their existing reward. The more they chose "A", the more rewards they received. Finally, participants were asked to write down the number of questions they had answered. In this study, the unethical behaviour was measured according to the number of A's chosen by the participants. That means that unethical behaviour has a value between 0 and 4.

3.2 | Results and discussion

3.2.1 | Materialism manipulation check

The independent-sample *t*-test results revealed that participants in the materialism priming group reported higher state materialism scores (N=30, M=56.70, SD=8.79) than those in the control group (N=29, M=51.62, SD=8.93), $t(57)=2.20$, $p=0.03$, Cohen's $d=0.57$, 95% CI=[0.46, 9.70], indicating that the manipulation was successful.

3.2.2 | The effect of materialism on unethical behaviour

The independent-sample *t*-test results showed that participants in the materialism priming group reported that they knew the answers to more questions ($t(57)=1.97$, M=1.10, SD=0.76) than those in the control group (M=0.72, SD=0.70), and the difference between the two groups was marginally significant ($p=0.053$, Cohen's $d=0.51$).

As expected, Study 2 demonstrated that the manipulation of materialism increased individuals' unethical behaviour. However, the effect of materialism on unethical behaviour was marginally significant. One limitation of this study was that participants might have falsely believed that they found an answer, although they did not. Such misreporting may not be equivalent to dishonest behaviour in a knowledge-based task (Gerlach et al., 2019). Therefore, to avoid the influence of some additional factors due to the knowledge-based task, Study 3 used a different and more objective method, a roll-a-die task, to measure unethical behaviour in order to enhance the reliability of our findings.

4 | STUDY 3

4.1 | Method

4.1.1 | Participants

The study was approved by our institutional ethics committee, and a total of 86 college students in Wuhan (a capital city of Hubei province in China) participated in this experiment and all participants provided informed content. After removing five participants who could not immerse themselves in the imagination task of manipulating materialism or ignored lab rules (e.g., taking phone calls during the experiment), the valid sample size was 81 (23 males; M_{age}=19.38, SD_{age}=0.80). All participants received a basic reward of 7 yuan and some additional rewards. The additional rewards depended on the number of dice in the roll dice task.

4.1.2 | Procedures and measures

The same experimental design, procedures, and materialism manipulation from Study 2 were used in Study 3.

State materialism

We used 9-item MVS developed by Richins (2004) and prefixed "at this moment" to the beginning of each item (e.g., At this moment, my life would be better if I owned certain things I don't have.). All items were scored on a 5-point scale (1=strongly disagree, 5=strongly agree), and after reverse scoring, the total score was used to represent the level of individuals' state materialism. In this study, the Cronbach's α was 0.71.

Unethical behaviour

After completing state materialism and some unrelated measures, participants engaged in a task that was ostensibly about luck (Gerlach et al., 2019; Lu et al., 2020). This task instructed participants to roll a die and self-report the outcome. Participants were told that the number of dice points reported would determine the additional rewards they received beyond their basic rewards (i.e., ¥1 for 1, ¥2 for 2, … ¥6 for 6).

4.2 | Results and discussion

4.2.1 | Materialism manipulation check

The independent-sample *t*-test results showed that participants in the materialism priming group reported higher

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state materialism scores (N=38, M=29.29, SD=5.08) than those in the control group (N=43, M=25.35, SD=4.62), $t(79)=3.66$, $p<0.001$, Cohen's $d=0.81$, 95% CI=[1.80, 6.09], indicating that the manipulation was successful.

4.2.2 | The effect of materialism on unethical behaviour

In terms of probability, the expected outcome of the dice-rolling task would be (1+2+3+4+5+6)/6=3.50, if no participant cheated. Therefore, following Lu et al. (2020), we compared the scores of the materialism priming group and control group to 3.5. The results showed that participants in the materialism priming group (M=4.18, SD=1.33) reported more dice points than 3.5 (t (37)=3.17, $p=0.003$, Cohen's $d=0.51$, 95% CI=[0.25, 1.12]), whereas the dice points reported by those in the control group (M=3.86, SD=1.70) were not significantly higher than 3.5 ($t(42)=1.39$, $p=0.17$, 95% CI=[−0.16, 0.88]). This result indicated that materialism priming increased cheating behaviour.

Studies 2 and 3 demonstrated the causal relationship between materialism and unethical behaviour but lacked an exploration of the mechanisms underlying this relationship. Thus, Study 4 investigated the mediating role of self-control in the relationship between the two.

5 | STUDY 4

5.1 | Method

5.1.1 | Participants

A total of 342 questionnaires were collected on Wenjuan.com. Wenjuan.com is an online crowdsourcing platform in mainland China that provides services equivalent to Amazon Mechanical Turk. The participants were aged between 17 and 55 years (M_{age}=22.58, SD=2.98), among whom 67.3% were female. Valid participants were randomly rewarded 1–3 yuan.

5.1.2 | Measures

Materialism

We used the 13-item Chinese version of MVS (Li & Guo, 2009), which was revised based on the original 18-item MVS (Richins & Dawson, 1992) and has been widely used in China. A sample item is "I usually buy only the things I need.". Participants responded on a 5-point scale (1=strongly disagree to 5=strongly agree). In this study, Cronbach's α was 0.77.

Self-control

We measured self-control with the Chinese version of self-control scale (Yi, 2014), which was revised based on the self-control scale developed by Tangney et al. (2004). A sample item is "I can resist temptation very well." Participants responded on a 5-point scale ranging from 1 (not at all) to 5 (very much). In this study, the Cronbach's α was 0.80.

Unethical behaviour

The measure of unethical behaviour was adapted from the Unethical Decision-Making Scale (Detert et al., 2008), in which individuals assess how likely they are to be unethical in a specific social context. We only selected four scenarios from the original scale because other scenarios were not completely consistent with the reality of Chinese university students' lives. Participants responded to each scenario on a 7-point scale (1=very unlikely to 7=very likely). The higher the score, the more likely it is that an individual will act unethically in a specific social context. Cronbach's α for this measure was 0.74.

Subjective social class (SSC)

Piff et al. (2012) found that higher social class predicted increased unethical behaviour. Therefore, in this study, we measured SSC as a control variable. Participants selected their perceived social class from five categories: lower class, lower-middle class, middle class, upper-middle class, and upper class.

5.2 | Results and discussion

5.2.1 | Common method biases test

Since all the data in this study were self-reported by participants at the same time, common method biases might have been present, the Harman single-factor test was conducted (Podsakoff et al., 2003) before the data analysis. The results showed that the eigenvalues of eight common factors were greater than 1, and the first common factor explained only 18.18% of the total variance, which was less than the critical standard of 40%, indicating that the common method bias in this study was not serious.

5.2.2 | Descriptive statistics and correlation analysis

As shown in Table 2, there was a significant negative correlation between materialism and self-control ($r=-0.31$, $p<0.001$), and the association between self-control and unethical behaviour was significant ($r=-0.26$, $p<0.001$). In addition, materialism positively correlated with unethical behaviour ($r=0.20$, $p<0.001$).

5.2.3 | Mediation effect of self-control

The mediation analysis was conducted using Process macro (Model 4) with 5000 bootstrapped resamples

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(Hayes, 2013). Gender, age, and SSC were used as control variables, and the results are shown in Figure 1. Consistent with our hypothesis, the total effect of materialism on unethical behaviour was significant (β=0.22, p<0.001, 95% CI=[0.56, 1.78]). Materialism negatively predicted self-control (β=−0.33, p<0.001, 95% CI=[−0.44, −0.20]), which in turn negatively predicted unethical behaviour (β=−0.20, p<0.001, 95% CI=[−0.31, −0.08]). The residual direct effect was also significant (β=0.16, p=0.005, 95% CI=[0.03, 0.28]). Self-control, therefore, played a mediating role in the link between materialism and unethical behaviour (indirect effect=0.07, 95% CI=[0.02, 0.12]), and the proportion of the mediation effect was 30%.

Study 4 confirmed the mediating role of self-control in the relationship between materialism and unethical behaviour. Specifically, the higher the individuals' level of materialism, the lower their self-control, and consequently, the more unethical behaviour they would exhibit. In Study 5, we adopted the manipulation-of-mediator design (Pirlott & MacKinnon, 2016) to further test the mediating role of self-control. Drawing on the designs of previous studies (Li et al., 2012; Wu et al., 2022), high materialists were assigned to different levels of self-control in Study 5 to examine whether different levels of self-control would affect their unethical behaviour. If inducing high materialists' self-control can reduce their unethical behaviour, we can conclude that it is self-control, not another variable, that accounts for the effect of materialism on unethical behaviour, which can provide a stronger ground for the mediation model.

6 | STUDY 5

6.1 | Method

6.1.1 | Participants

Since Study 4 indicated that high materialists tend to have a lower level of self-control, thus leading to more unethical behaviours, in Study 5, we further examined whether priming self-control of high materialists could reduce their unethical behaviour. Therefore, only high materialists were invited to participate in our experiment. To select the participants we need, a total of 264 undergraduates from a university in Wuhan, China completed the Chinese version of the Material Value Scale (Li & Guo, 2009) online and their average scores per item on this scale (13 items, 5-point scoring) were calculated. Thereby, 124 undergraduates whose average score per item was greater than 3 were regarded as high materialists, and 96 of them volunteered to participate in the subsequent experiment. Six participants were removed because they did not follow the instructions of the experiment task, specifically, they wrote far fewer than 120 words in the manipulation of self-control. Our final sample consisted of 90 undergraduates (23 males;

TABLE 2 Descriptive statistics and correlation analysis in Study 4.

Variables	*M*	SD	1	2	3	4	5
1. Gender[a]	1.67	0.47	1				
2. Age	22.58	2.98	−0.18***	1			
3. Subjective social class	2.40	0.80	0.17**	−0.03	1		
4. Materialism	40.55	6.19	0.13*	−0.14**	0.04	1	
5. Self-control	37.57	6.24	0.01	0.15**	0.12*	−0.33***	1
6. Unethical behaviour	13.44	5.24	−0.18**	−0.02	−0.11*	0.20***	−0.26***

[a]Male=1, female=2.
*p<0.05, **p<0.01, ***p<0.001.

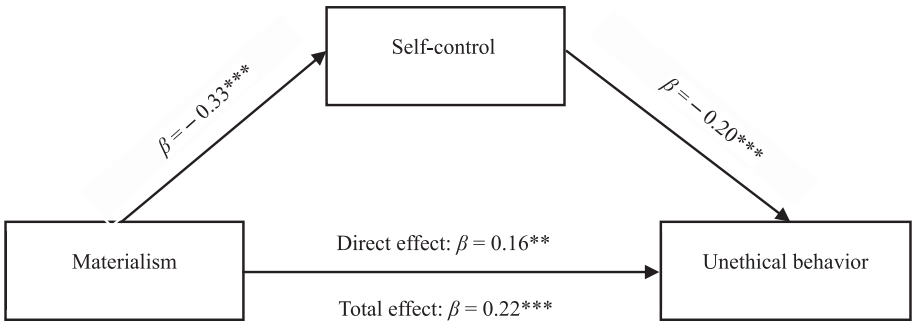


FIGURE 1 Mediation model. All variables were standardized. **p<0.01, ***p<0.001.

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M_{age} = 19.43, SD_{age} = 1.31). After the experiment, all participants received 6 yuan as a token of appreciation.

6.1.2 | Experimental design and procedures

The experiment was a single-factor between-subjects design. After arriving at the laboratory, participants first completed a recall writing task designed to manipulate their level of self-control. They then responded to a question that measured state self-control. Finally, the participants completed the Unethical Decision-Making Scale and Trait Self-Control Scale.

6.1.3 | Materials and measures

Materialism

The measure of materialism was the same as in Study 4. In this study, the Cronbach's α was 0.82.

The manipulation of self-control

We used a recall writing task to manipulate self-control, adapted from Kokkoris et al. (2019) and Ge and Hou (2021). Specifically, participants in the high self-control group were asked to recall a particular event in which they wanted to do something that conflicted with their goal but ultimately chose the goal (e.g., want things: eat chocolate vs. goal: lose weight). The low self-control group also recalled a specific event where they wanted to do something that conflicted with their goal and eventually gave up the goal. Finally, both groups wrote down the specific event in no less than 120 words.

State self-control

Following Ge and Hou (2021), we used one item to check the manipulation of state self-control. Participants answered "What level of your self-control does the event you just recalled and wrote down reflect" and responded on a 7-point Likert scale (1 = low self-control to 7 = high self-control).

Unethical behaviour

The measure of unethical behaviour was similar to that in Study 4, in which participants assessed how likely they were to be unethical in a specific social context. Considering the duration of the priming effect of the experiment and to minimize the potential effect of participants' fatigue on the sensitivity of the dependent variable, we only selected two scenarios from the four scenarios used in Study 4 which are related to the study and most suitable for students. Participants responded to each scenario on a 7-point scale (1 = very unlikely to 7 = very likely). A higher total score in both scenarios represents a higher level of unethical behaviour. In this study, the reliability of this two-item measure (Spearman-Brown coefficient; Eisinga et al., 2013) was 0.63.

Trait self-control

Participants' own traits of self-control may influence their unethical behaviour. Therefore, trait self-control was treated as the control variable in our study. The measurement was the same as in Study 4. In this study, Cronbach's α was 0.76.

Subjective social class

The measurement was the same as in Study 4.

6.2 | Results and discussion

6.2.1 | Self-control manipulation check

The independent-sample *t*-test results revealed that participants in the high self-control condition had higher state self-control scores (N = 43, M = 5.02, SD = 0.94) than those in the low self-control condition (N = 47, M = 2.57, SD = 1.06), $t(88) = 11.57$, $p < 0.001$, Cohen's $d = 2.44$, 95% CI = [2.03, 2.87], indicating that the manipulation was successful.

6.2.2 | The effect of self-control on unethical behaviour

The results of independent-sample *t*-test showed that participants in the high self-control group (M = 7.60, SD = 2.90) reported lower scores on unethical behaviour than those in the low self-control group (M = 9.47, SD = 3.36), $t(88) = -2.81$, $p = 0.006$, Cohen's $d = 0.59$, 95% CI = [−3.18, −0.54].

Further, trait self-control was included as a control variable in ANCOVA to exclude its potential effect on the study results. Meanwhile, gender, age, and subjective social class were also controlled. The results showed that there was still a significant difference between the scores of the two groups on unethical behaviour, $F(1, 84) = 7.37$, $p = 0.008$, $\eta^2 = 0.08$.

Study 5 examined the influence of self-control on unethical behaviour of high materialists by assigning them to different levels of self-control. As expected, activating high materialists' self-control reduced their unethical behaviour, which further verified the mediating role of self-control between materialism and unethical behaviour.

7 | GENERAL DISCUSSION

The current research examined the effect of materialism on unethical behaviour and its psychological mechanisms across five studies in Chinese samples. In Study 1, we used data from the Sixth World Values Survey and found that materialism was an positive predictor of a wide range of unethical behaviours, such as fare evasion, theft, tax evasion, bribery, domestic violence, and violence against others.

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Studies 2 and 3 manipulated materialism to further reveal the causal relationship between materialism and dishonest behaviour. In Study 4, through a questionnaire survey, we found that materialism positively predicted unethical behaviour in specific social contexts and self-control played a mediating role in this relationship. Study 5 showed that activating high materialists' self-control reduced their unethical behaviour, further confirming the mediating role of self-control.

7.1 | Contributions and implications

This research makes some theoretical and methodical contributions. Theoretically, first, this research expands the study of the mechanisms underlying the relationship between materialism and unethical behaviour by revealing the mediating role of self-control. Very few studies have found the mediating role of machiavellianism (Du, 2018) and willingness to cheat (Koul, 2012). Our research demonstrates that self-control is also an important mediator to explain how materialism influence unethical behaviour. According to the hot/cool system, human behaviour is influenced by cold and hot systems. When the hot system is dominant, individuals exercise less conscious self-control and are therefore relatively prone to impulsive behaviour (de Ridder et al., 2011; Metcalfe & Mischel, 1999). Materialists tend to be impulsive (Li et al., 2016; Roberts & Pirog, 2013; Rose, 2007) and are more likely to act under the hot system which undermines their effort of self-control, with less regard for moral principles, and thus exhibit unethical behaviour. Obviously, our findings support the hot/cool system model of self-control.

Also, our study contributes to understanding antecedents of unethical behaviour from the perspective of values. Previous studies have demonstrated the positive link between self-enhancement values (power, achievement, and hedonism) and unethical behaviour (Arciniega et al., 2019; Feldman et al., 2015; Ring et al., 2020). Materialism is also a kind of self-enhancement value (Burroughs & Rindfleisch, 2002). Consistent with the findings of existing literature (e.g., Chowdhury & Fernando, 2013; Gentina et al., 2018; Koul, 2012; Liang et al., 2016; Lu & Lu, 2010), we also found its positive effect on unethical behaviour.

In addition, this research enriches the studies about the consequences of materialism. Materialism, to some extent, stimulates people's economic or work motivation and then promotes the improvement of residents' quality of life (Sirgy et al., 2013), and it can also be used as a compensation strategy for a personal sense of insecurity (e.g., low self-esteem) in the short term (Li et al., 2018; Shrum et al., 2014). However, there have been a large number of studies demonstrating that, in the long run, materialism has negative effects on both individual and societal well-being (see reviews, Dittmar & Isham, 2022; Kasser, 2016; Moldes & Ku, 2020). Our study reveals the negative effect of materialism on unethical behaviour, which contributes to understanding how attitudes and values that are prevailing in society might negatively influence the societal well-being.

Methodically, to the best of our knowledge, it is the first study to systematically explore the effect of materialism on unethical behaviour with multiple methods, samples, and measurement indicators of unethical behaviour. Compared with the limited existing studies which are mostly correlational and only include a single indicator of unethical behaviour, this study provides more convincing evidence regarding the positive link between materialism and unethical behaviour. In addition, in our study, we manipulate mediator by "manipulation-of-mediator design" (Li et al., 2012; Pirlott & MacKinnon, 2016; Wu et al., 2022) to enhance the robustness of mediation model relative to some studies that used "measurement-of-mediation designs" (e.g., study 4 in our studies). Overall, our multi-method design enhances the robustness of mediation model as well as the causal relationship between materialism and unethical behaviour.

7.2 | Limitations and future research directions

Some limitations of this study need to be addressed in future research. First, we only used an imagination-based paradigm to manipulate materialism. In future research, multiple priming approaches, such as the descrambling task (Bauer et al., 2012) and picture task (Vázquez et al., 2021), could be used to investigate the effect of materialism on unethical behaviour. Second, our research only explored the mediating role of self-control. It is advisable for future research to further explore the boundary conditions between materialism and unethical behaviour. For example, awe, as a type of self-transcendence emotion, can motivate individuals to engage in many positive behaviours (Keltner & Haidt, 2003; Piff et al., 2015; Prade & Saroglou, 2016). Previous research has found that awe plays an important role in reducing individuals' desire for material things and money (Jiang et al., 2018; Tian & Lu, 2015). Therefore, it would be valuable to examine the moderating role of awe in the future.

8 | CONCLUSION

This research examined the effect of materialism on unethical behaviour and the mediating role of self-control in this relationship by conducting five muti-method studies with Chinese samples. The results showed that materialism had a positive effect on unethical behaviour and self-control mediated this relationship, indicating that

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materialists tend to have lower levels of self-control, thus they are more likely to engage in unethical behaviour.

AUTHOR CONTRIBUTIONS

Lei Yue: Conceptualization; data curation; formal analysis; investigation; methodology; writing – original draft; writing – review and editing. **Xiangli Zeng:** Conceptualization; investigation; writing – original draft. **Jing Li:** Conceptualization; data curation; funding acquisition; methodology; project administration; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare there are no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data and materials for this study are available through the Open Science Framework page: https://osf.io/vcx34/.

ETHICS STATEMENT

All studies were conducted in accordance with ethical guidelines in psychology and obtained ethical approval from the Ethics Committee and the Institutional Review Board of Central China Normal University.

RESEARCH MATERIALS AVAILABILITY STATEMENT

The materials for this study are available through the Open Science Framework page: https://osf.io/vcx34/.

PRE-REGISTRATION STATEMENT

The studies presented in this paper were not preregistered.

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ENDNOTE

[1] The World Values Survey (WVS) is the largest international research program on non-commercial empirical time series. It was started in 1981 by renowned scholar Ronald Inglehart and his team. The WVS contains social, cultural, political, economic, and religious values of people in the world and is conducted globally every 5 years. The Chinese data from the sixth World Values Survey (WVS Wave 6) was conducted in 2012.

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